

Technology Stack

SMARTBRIDGE / SKILL WALL

Date: 29 September 2026

Team ID: SWTID-2026-3226

Project Name: FitBuddy - AI Fitness Plan Generator using Gemini Models

Maximum Marks: 2 Marks

Define Your Technology Stack

Identify the programming languages, frameworks, data storage, front-end and back-end tools, and APIs used to build the solution. The selected stack supports maintainability and FitBuddy's AI-powered fitness-planning requirements.

The table below lists the technologies selected for each architecture component and their purpose.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML + CSS + JavaScript	Provides a responsive interface for entering fitness goals, activity level, schedule, equipment, and preferences, then viewing the generated plan.
2	Backend / Server-Side	Python + FastAPI	Provides REST endpoints, input validation, request routing, plan-generation orchestration, and server-side integration with Gemini.
3	Database / Data Storage	SQLite (MVP)	Lightweight local persistence for user preferences and generated-plan history, supporting the profile and history features without a separate database service.
4	Cloud / Hosting / Deployment	Uvicorn + Local Deployment	Runs the FastAPI application locally for development, testing, and demonstration; production hosting can be selected when deployment needs are defined.
5	Version Control & CI/CD	Git + GitHub	Stores source code and project deliverables, tracks changes, supports team collaboration, and can host automated checks.
6	Third-Party APIs / Other Tools	Google Gemini API	Generates personalized fitness plans from validated user goals and preferences, including workout schedules, exercise guidance, progression, and optional general nutrition suggestions.

Selected Stack Summary

Frontend: HTML, CSS, JavaScript | **Backend:** Python, FastAPI | **AI:** Google Gemini API

Deployment: Uvicorn / local development | **Version Control:** Git / GitHub | **Data:** SQLite for profile and plan history